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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 24th January Evening Shift

JEE Main Question Paper with Answer Key

24 Jan 2023	90	17	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	22 questions
Physics	29 questions
Out of syllabus	17 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

The number of s-electrons present in an ion with 55 protons in its unipositive state is

(A) 10 (B) 9 (C) 12 (D) 8

Open solution

Q2. MCQ (Single Correct Answer)

Find out the major products from the following reactions.



Open solution

(A)



(B)



(C)



(D)



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Q3. MCQ (Single Correct Answer)

What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : $N_2; N_2^+; O_2; O_2^+$?

(A) 0, 1, 0, 1 (B) 2, 1, 0, 1 (C) 0, 1, 2, 1 (D) 2, 1, 2, 1

Open solution

Q4. MCQ (Single Correct Answer)

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Benzene is more stable than hypothetical cyclohexatriene

Reason R : The delocalised π electron cloud is attracted more strongly by nuclei of carbon atoms.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is true but R is false

(B) Both A and R are correct and R is the correct explanation of A

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) A is false but R is true

Open solution

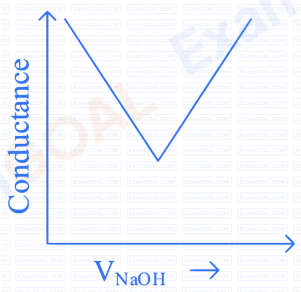
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Q5. MCQ (Single Correct Answer)

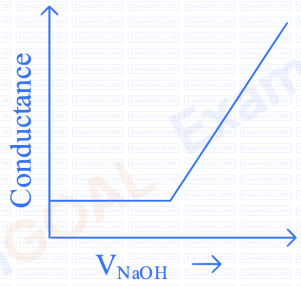
Open solution

Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.

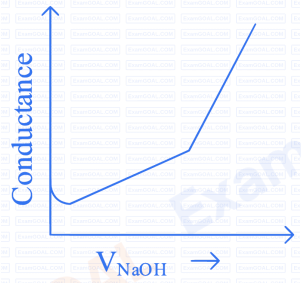
(A)



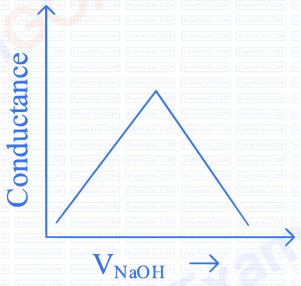
(B)



(C)



(D)



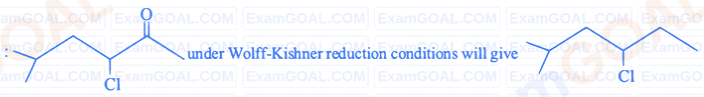
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Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : 

Statement II : 

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

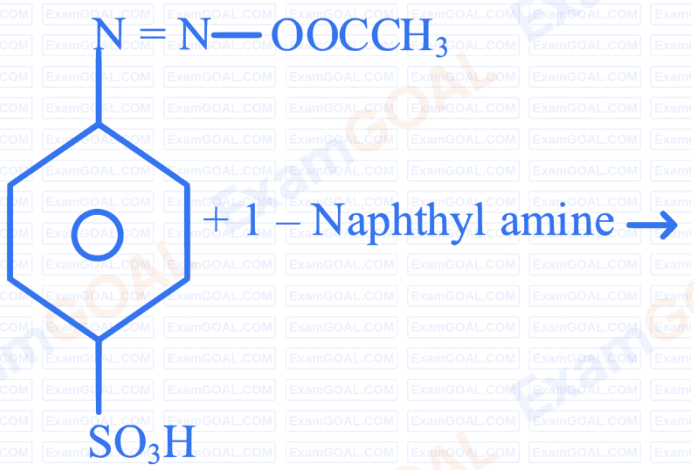
(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q7. MCQ (Single Correct Answer)

Open solution

Choose the correct colour of the product for the following reaction.



(A) White (B) Yellow (C) Red (D) Blue

Q8. MCQ (Single Correct Answer)

Open solution

Which one amongst the following are good oxidizing agents?

A. Sm^{2+}

B. Ce^{2+}

C. Ce^{4+}

D. Tb^{4+}

Choose the most appropriate answer from the options given below :

(A) C and D only (B) A and B only (C) D only (D) C only

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Q9. MCQ (Single Correct Answer)

A student has studied the decomposition of a gas AB_3 at 25°C. He obtained the following data.

p (mm Hg)	50	100	200	400
relative $t_{1/2}$ (s)	4	2	1	0.5

The order of the reaction is

(A) 2 (B) 0.5 (C) 1 (D) 0 (zero)

Open solution

Q10. MCQ (Single Correct Answer)

The hybridization and magnetic behaviour of cobalt ion in $[Co(NH_3)_6]^{3+}$ complex, respectively is :

(A) sp^3d^2 and paramagnetic
(B) sp^3d^2 and diamagnetic
(C) d^2sp^3 and diamagnetic
(D) d^2sp^3 and paramagnetic

Open solution

Q11. MCQ (Single Correct Answer)

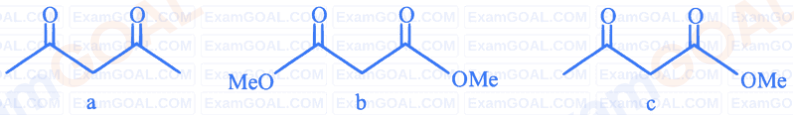
$K_2Cr_2O_7$ paper acidified with dilute H_2SO_4 turns green when exposed to :

(A) Sulphur dioxide
(B) Sulphur trioxide
(C) Hydrogen sulphide
(D) Carbon dioxide

Open solution

Q12. MCQ (Single Correct Answer)

Which will undergo deprotonation most readily in the basic medium?



(A) a only (B) Both a and c (C) c only (D) b only

Open solution

Q13. MCQ (Single Correct Answer)

Which of the following cannot be explained by crystal field theory?

(A) The order of spectrochemical series
(B) Stability of metal complexes
(C) Magnetic properties of transition metal complexes
(D) Colour of metal complexes

Open solution

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Q14. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I : Pure Aniline and other arylamines are usually colourless.

Statement II : Arylamines get coloured on storage due to atmospheric reduction.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Q15. Numerical

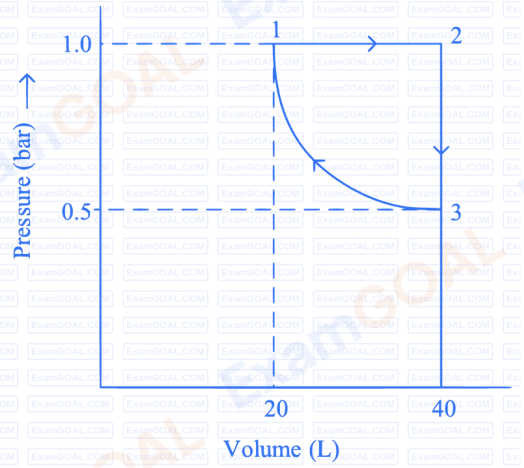
Open solution

Total number of tripeptides possible by mixing of valine and proline is _____

Q16. Numerical

Open solution

One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is _____ J (nearest integer)



Given ; $\log 2 = 0.3$

$\ln 10 = 2.3$

Q17. Numerical

Open solution

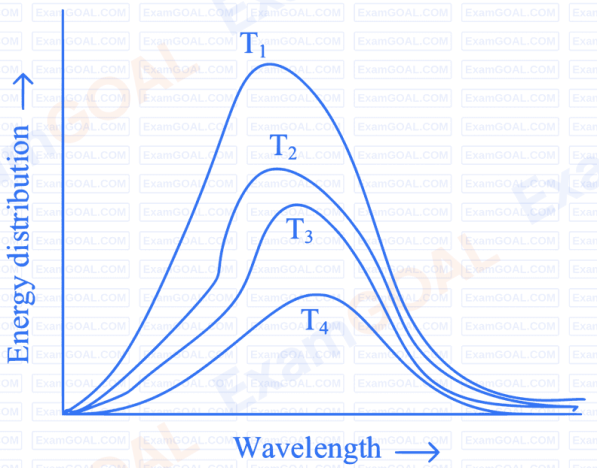
Maximum number of isomeric monochloro derivatives which can be obtained from 2,2,5,5-tetramethylhexane by chlorination is _____

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Q18. Numerical

Following figure shows spectrum of an ideal black body at four different temperatures. The number of correct statement/s from the following is _____.

Open solution



A. $T_4 > T_3 > T_2 > T_1$

B. The black body consists of particles performing simple harmonic motion.

C. The peak of the spectrum shifts to shorter wavelength as temperature increases.

D. $\frac{T_1}{v_1} = \frac{T_2}{v_2} = \frac{T_3}{v_3} \neq$ constant

E. The given spectrum could be explained using quantisation of energy.

Q19. Numerical

Sum of π-bonds present in peroxodisulphuric acid and pyrosulphuric acid is _____

Open solution

Q20. Numerical

The number of units, which are used to express concentration of solutions from the following is _____

Mass percent, Mole, Mole fraction, Molarity, ppm, Molality

Open solution

Q21. Numerical

The total pressure observed by mixing two liquids A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively.

The total pressure becomes 410 mm Hg if the mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of pure A is _____ mm Hg. (Nearest integer)

Consider the liquids and solutions behave ideally.

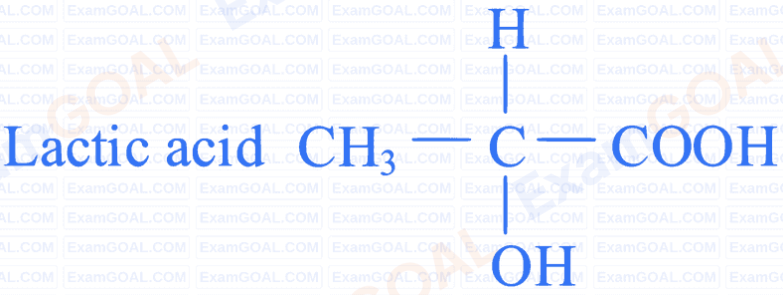
Open solution

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Q22. Numerical

If the pKa of lactic acid is 5, then the pH of 0.005 M calcium lactate solution at 25°C is _____ $\times 10^{-1}$ (Nearest integer)

Open solution

Lactic acid 

Mathematics 22 questions

Q23. MCQ (Single Correct Answer)

Let $f(x)$ be a function such that $f(x+y)=f(x).f(y)$ for all $x, y \in \mathbb{N}$. If $f(1)=3$ and $\sum_{k=1}^{n} f(k)=3279$, then the value of n is

(A) 9 (B) 7 (C) 6 (D) 8

Open solution

Q24. MCQ (Single Correct Answer)

The number of real solutions of the equation $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, is

(A) 3 (B) 4 (C) 0 (D) 2

Open solution

Q25. MCQ (Single Correct Answer)

The number of integers, greater than 7000 that can be formed, using the digits 3, 5, 6, 7, 8 without repetition is :

(A) 48 (B) 120 (C) 168 (D) 220

Open solution

Q26. MCQ (Single Correct Answer)

Let $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ and $\vec{\beta}=\hat{i}+2\hat{j}-4\hat{k}$. Let $\vec{\beta_1}$ be parallel to $\vec{\alpha}$ and $\vec{\beta_2}$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta}=\vec{\beta_1}+\vec{\beta_2}$, then the value of $5\vec{\beta_2}.\left(\hat{i}+\hat{j}+\hat{k}\right)$ is :

(A) 9 (B) 7 (C) 6 (D) 11

Open solution

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Q27. MCQ (Single Correct Answer)

If the system of equations

$x + 2y + 3z = 3$

$4x + 3y - 4z = 4$

$8x + 4y - \lambda z = 9 + \mu$

has infinitely many solutions, then the ordered pair (λ, μ) is equal to :

(A) $\left(\frac{72}{5}, \frac{21}{5}\right)$ (B) $\left(-\frac{72}{5}, -\frac{21}{5}\right)$

(C) $\left(-\frac{72}{5}, \frac{21}{5}\right)$ (D) $\left(\frac{72}{5}, -\frac{21}{5}\right)$

Open solution

Q28. MCQ (Single Correct Answer)

The set of all values of a for which $\lim\limits_{x \to a}([x-5] - [2x+2]) = 0$, where $[\alpha]$ denotes the greatest integer less than or equal to α is equal to

(A) $[-7.5, -6.5]$ (B) $(-7.5, -6.5]$ (C) $[-7.5, -6.5)$ (D) $(-7.5, -6.5)$

Open solution

Q29. MCQ (Single Correct Answer)

The locus of the mid points of the chords of the circle $C_1 : (x-4)^2 + (y-5)^2 = 4$ which subtend an angle θ_i at the centre of the circle C_1, is a circle of radius r_i. If $\theta_1 = \frac{\pi}{3}, \theta_3 = \frac{2\pi}{3}$ and $r_1^2 = r_2^2 + r_3^2$, then θ_2 is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{6}$

Open solution

Q30. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation $(x^2 - 3y^2)dx + 3xy\,dy = 0, y(1) = 1$. Then $6y^2(e)$ is equal to

(A) $\frac{3}{2}e^2$ (B) $3e^2$ (C) e^2 (D) $2e^2$

Open solution

Q31. MCQ (Single Correct Answer)

If $f(x) = \frac{2^{2x}}{2^{2x}+2}, x \in \mathbb{R}$, then $f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + \ldots + f\left(\frac{2022}{2023}\right)$ is equal to

(A) 2011 (B) 2010 (C) 1010 (D) 1011

Open solution

Q32. MCQ (Single Correct Answer)

The number of square matrices of order 5 with entries from the set {0, 1}, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is :

(A) 125 (B) 150 (C) 225 (D) 120

Open solution

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Q33. MCQ (Single Correct Answer)

Let the six numbers $a_1, a_2, a_3, a_4, a_5, a_6$, be in A.P. and $a_1 + a_3 = 10$. If the mean of these six numbers is $\frac{19}{2}$ and their variance is σ^2, then $8\sigma^2$ is equal to :

(A) 220 (B) 210 (C) 105 (D) 200

Open solution

Q34. MCQ (Single Correct Answer)

The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is

(A) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (B) $-\frac{1}{2}\left(\sqrt{3}-i\right)$

(C) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(\sqrt{3}+i\right)$

Open solution

Q35. MCQ (Single Correct Answer)

If $f(x) = x^3 - x^2 f'(1) + x f''(2) - f'''(3), x \in \mathbb{R}$, then

(A) $2f(0) - f(1) + f(3) = f(2)$ (B) $f(1) + f(2) + f(3) = f(0)$

(C) $f(3) - f(2) = f(1)$ (D) $3f(1) + f(2) = f(3)$

Open solution

Q36. MCQ (Single Correct Answer)

$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) 2π

Open solution

Q37. Numerical

The minimum number of elements that must be added to the relation R = {(a, b), (b, c), (b, d)} on the set {a, b, c, d} so that it is an equivalence relation, is _____.

Open solution

Q38. Numerical

If the shortest distance between the lines $\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is 6, then the square of sum of all possible values of λ is :

Open solution

Q39. Numerical

Let $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}, \vec{a}.\vec{c} = 7, 2\vec{b}.\vec{c} + 43 = 0, \vec{a} \times \vec{c} = \vec{b} \times \vec{c}$. Then $\left|\vec{a}.\vec{b}\right|$ is equal to :

Open solution

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Q40. Numerical

Let the sum of the coefficients of the first three terms in the expansion of $\left(x-\frac{3}{x^2}\right)^n, x \neq 0. n \in \mathbb{N}$, be 376. Then the coefficient of x^4 is _____.

Open solution

Q41. Numerical

The equations of the sides AB, BC and CA of a triangle ABC are : $2x+y=0, x+py=21a, (a \neq 0)$ and $x-y=3$ respectively. Let P(2, a) be the centroid of ΔABC. Then $(BC)^2$ is equal to _____.

Open solution

Q42. Numerical

Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x)>0$ and $f(x)+\int_0^x f(t)\sqrt{1-(\log_e f(t))^2}dt=e, \forall x \in \left[0, \frac{\pi}{2}\right]$. Then $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to _____.

Open solution

Q43. Numerical

If the area of the region bounded by the curves $y^2-2y=-x, x+y=0$ is A, then 8 A is equal to _____

Open solution

Q44. Numerical

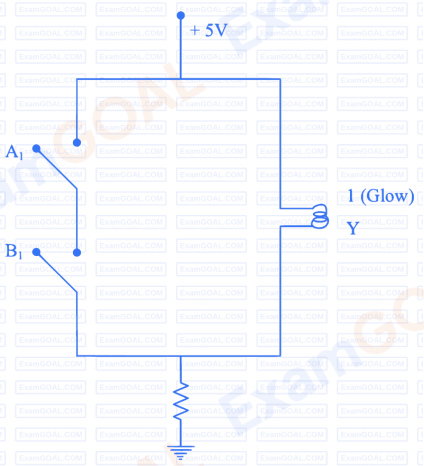
Three urns A, B and C contain 4 red, 6 black; 5 red, 5 black; and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola $y^2=\lambda x$ with one vertex at the vertex of the parabola, is :

Open solution

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Physics 29 questions

Q45. MCQ (Single Correct Answer)



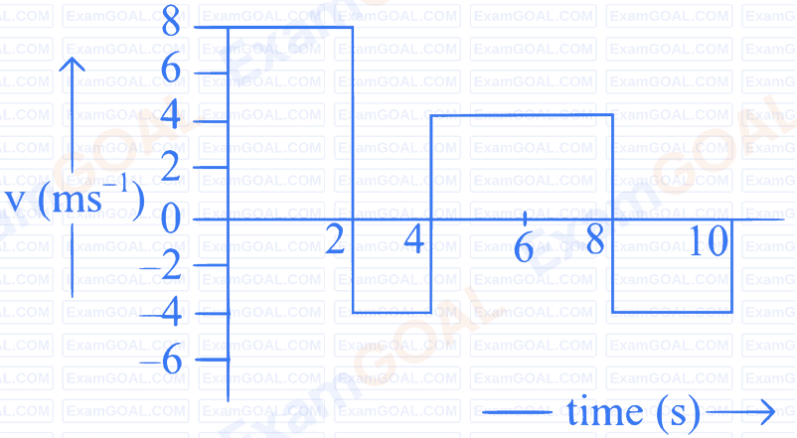
Open solution

The logic gate equivalent to the given circuit diagram is :

(A) NOR (B) OR (C) NAND (D) AND

Q46. MCQ (Single Correct Answer)

The velocity time graph of a body moving in a straight line is shown in the figure.



Open solution

The ratio of displacement to distance travelled by the body in time 0 to 10s is :

(A) 1 : 1 (B) 1 : 3 (C) 1 : 4 (D) 1 : 2

Q47. MCQ (Single Correct Answer)

A long solenoid is formed by winding 70 turns cm^{-1}. If 2.0 A current flows, then the magnetic field produced inside the solenoid is _____ ($\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1})

(A) 88×10^{-4} T (B) 1232×10^{-4} T (C) 176×10^{-4} T (D) 352×10^{-4} T

Open solution

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Q48. MCQ (Single Correct Answer)

A cell of emf 90 V is connected across series combination of two resistors each of 100Ω resistance. A voltmeter of resistance 400Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be :

Open solution

(A) 40 V (B) 90 V (C) 45 V (D) 80 V

Q49. MCQ (Single Correct Answer)

An α-particle, a proton and an electron have the same kinetic energy. Which one of the following is correct in case of their de-Broglie wavelength:

Open solution

(A) $\lambda_\alpha > \lambda_p < \lambda_e$ (B) $\lambda_\alpha > \lambda_p > \lambda_e$

(C) $\lambda_\alpha = \lambda_p = \lambda_e$ (D) $\lambda_\alpha < \lambda_p < \lambda_e$

Q50. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : A pendulum clock when taken to Mount Everest becomes fast.

Reason R : The value of g (acceleration due to gravity) is less at Mount Everest than its value on the surface of earth.

In the light of the above statements, choose the most appropriate answer from the options given below

Open solution

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) A is correct but R is not correct

(C) Both A and R are correct and R is the correct explanation of A

(D) A is not correct but R is correct

Q51. MCQ (Single Correct Answer)

Let γ_1 be the ratio of molar specific heat at constant pressure and molar specific heat at constant volume of a monoatomic gas and γ_2 be the similar ratio of diatomic gas. Considering the diatomic gas molecule as a rigid rotator, the ratio, $\frac{\gamma_1}{\gamma_2}$ is :

Open solution

(A) $\frac{35}{27}$ (B) $\frac{25}{21}$ (C) $\frac{21}{25}$ (D) $\frac{27}{35}$

Q52. MCQ (Single Correct Answer)

When a beam of white light is allowed to pass through convex lens parallel to principal axis, the different colours of light converge at different points on the principal axis after refraction. This is called :

Open solution

(A) Spherical aberration (B) Scattering

(C) Polarisation (D) Chromatic aberration

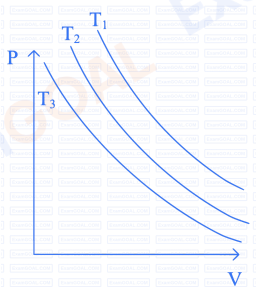
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Q53. MCQ (Single Correct Answer)

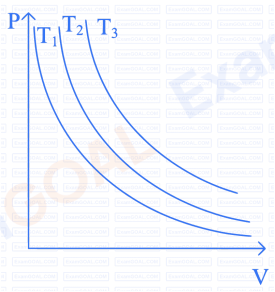
Open solution

In an Isothermal change, the change in pressure and volume of a gas can be represented for three different temperatures; $T_3 > T_2 > T_1$ as :

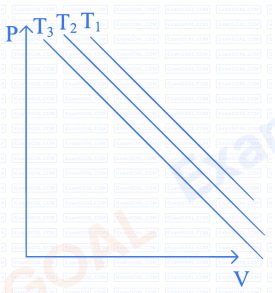
(A)



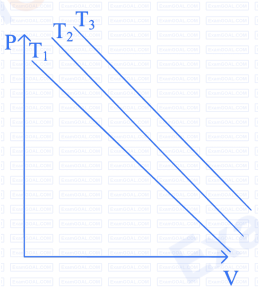
(B)



(C)



(D)



Q54. MCQ (Single Correct Answer)

Open solution

The electric field and magnetic field components of an electromagnetic wave going through vacuum is described by

$E_x = E_o \sin(kz - \omega t)$

$B_y = B_o \sin(kz - \omega t)$

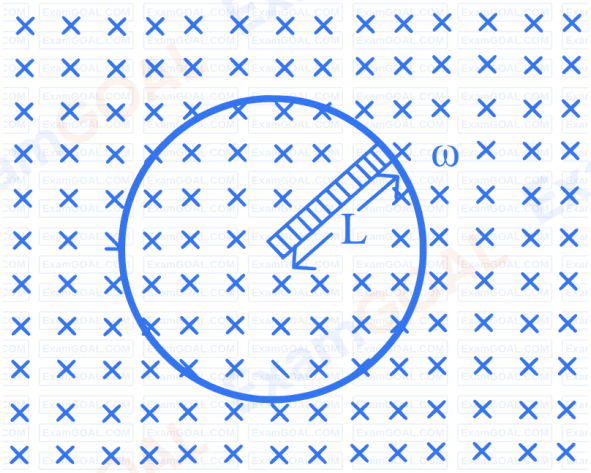
Then the correct relation between E_0 and B_0 is given by

(A) $E_o B_o = \omega k$ (B) $E_0 = kB_0$ (C) $kE_0 = \omega B_0$ (D) $\omega E_0 = kB_0$

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Q55. MCQ (Single Correct Answer)

A metallic rod of length 'L' is rotated with an angular speed of 'ω' normal to a uniform magnetic field 'B' about an axis passing through one end of rod as shown in figure. The induced emf will be :



Open solution

(A) $\frac{1}{2}B^2L^2\omega$ (B) $\frac{1}{2}BL^2\omega$ (C) $\frac{1}{4}BL^2\omega$ (D) $\frac{1}{4}B^2L\omega$

Q56. MCQ (Single Correct Answer)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Steel is used in the construction of buildings and bridges.

Reason R : Steel is more elastic and its elastic limit is high.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct and R is the correct explanation of A

(B) A is correct but R is not correct

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) A is not correct but R is correct

Open solution

Q57. MCQ (Single Correct Answer)

If the distance of the earth from Sun is 1.5×10^6 km. Then the distance of an imaginary planet from Sun, if its period of revolution is 2.83 years is :

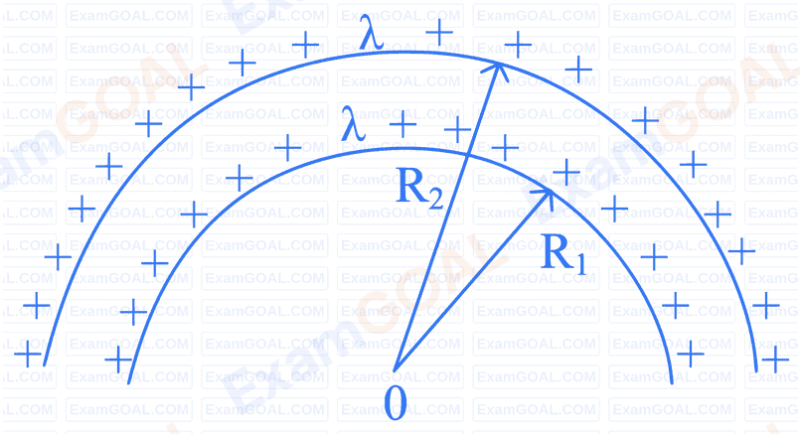
(A) 6×10^6 km (B) 3×10^7 km (C) 6×10^7 km (D) 3×10^6 km

Open solution

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Q58. MCQ (Single Correct Answer)

The electric potential at the centre of two concentric half rings of radii R_1 and R_2, having same linear charge density λ is :



(A) $\frac{\lambda}{2\epsilon_0}$ (B) $\frac{\lambda}{\epsilon_0}$ (C) $\frac{2\lambda}{\epsilon_0}$ (D) $\frac{\lambda}{4\epsilon_0}$

Open solution

Q59. MCQ (Single Correct Answer)

Given below are two statements:

Statement I : Acceleration due to earth's gravity decreases as you go 'up' or 'down' from earth's surface.

Statement II : Acceleration due to earth's gravity is same at a height 'h' and depth 'd' from earth's surface, if h = d.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Statement I is correct but statement II is incorrect

(B) Both Statement I and II are correct

(C) Statement I is incorrect but statement II is correct

(D) Both Statement I and Statement II are incorrect

Open solution

Q60. MCQ (Single Correct Answer)

A photon is emitted in transition from n = 4 to n = 1 level in hydrogen atom. The corresponding wavelength for this transition is (given, h = 4×10^{-15} eVs) :

(A) 99.3 nm (B) 94.1 nm (C) 974 nm (D) 941 nm

Open solution

Q61. MCQ (Single Correct Answer)

The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as $\nu = r^a \rho^b s^c$. The values of a, b and c respectively are

(A) $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (B) $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

(C) $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Open solution

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Q62. MCQ (Single Correct Answer)

If two vectors $\overrightarrow{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\overrightarrow{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be :

(A) -1 (B) 3 (C) 1 (D) 2

Open solution

Q63. MCQ (Single Correct Answer)

A body of mass 200g is tied to a spring of spring constant 12.5 N/m, while the other end of spring is fixed at point O. If the body moves about O in a circular path on a smooth horizontal surface with constant angular speed 5 rad/s, then the ratio of extension in the spring to its natural length will be :

(A) 1 : 2 (B) 2 : 3 (C) 2 : 5 (D) 1 : 1

Open solution

Q64. Numerical

A convex lens of refractive index 1.5 and focal length 18cm in air is immersed in water. The change in focal length of the lens will be _____ cm.

(Given refractive index of water $= \frac{4}{3}$)

Open solution

Q65. Numerical

A body of mass 1kg begins to move under the action of a time dependent force $\overrightarrow{F} = \left(t\hat{i} + 3t^2\,\hat{j}\right)$ N, where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y axis. The power developed by above force, at the time t = 2s, will be _____ W.

Open solution

Q66. Numerical

The energy released per fission of nucleus of ^{240}X is 200 MeV. The energy released if all the atoms in 120g of pure ^{240}X undergo fission is _____ $\times 10^{25}$ MeV.

(Given $\mathrm{N_A} = 6 \times 10^{23}$)

Open solution

Q67. Numerical

A uniform solid cylinder with radius R and length L has moment of inertia I_1, about the axis of the cylinder. A concentric solid cylinder of radius $R' = \frac{R}{2}$ and length $L' = \frac{L}{2}$ is carved out of the original cylinder. If I_2 is the moment of inertia of the carved out portion of the cylinder then $\frac{I_1}{I_2} =$ _____.

(Both I_1 and I_2 are about the axis of the cylinder)

Open solution

Q68. Numerical

If a copper wire is stretched to increase its length by 20%. The percentage increase in resistance of the wire is _____%.

Open solution

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Q69. Numerical

A parallel plate capacitor with air between the plates has a capacitance of 15pF. The separation between the plates becomes twice and the space between them is filled with a medium of dielectric constant 3.5. Then the capacitance becomes $\frac{x}{4}$ pF. The value of x is _____.

Open solution

Q70. Numerical

A single turn current loop in the shape of a right angle triangle with sides 5 cm, 12 cm, 13 cm is carrying a current of 2 A. The loop is in a uniform magnetic field of magnitude 0.75 T whose direction is parallel to the current in the 13 cm side of the loop. The magnitude of the magnetic force on the 5 cm side will be $\frac{x}{130}$ N. The value of x is _____.

Open solution

Q71. Numerical

A Spherical ball of radius 1mm and density 10.5 g/cc is dropped in glycerine of coefficient of viscosity 9.8 poise and density 1.5 g/cc. Viscous force on the ball when it attains constant velocity is 3696×10^{-x} N. The value of x is _____.

(Given, g = 9.8 m/s^2 and $\pi = \frac{22}{7}$)

Open solution

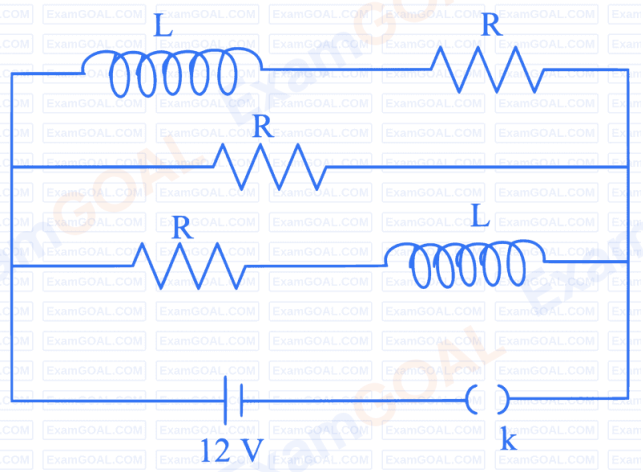
Q72. Numerical

A mass m attached to free end of a spring executes SHM with a period of 1s. If the mass is increased by 3 kg the period of the oscillation increases by one second, the value of mass m is _____ kg.

Open solution

Q73. Numerical

Three identical resistors with resistance R = 12Ω and two identical inductors with self inductance L = 5 mH are connected to an ideal battery with emf of 12 V as shown in figure. The current through the battery long after the switch has been closed will be _____ A.



Open solution

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Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q74. MCQ (Single Correct Answer) · **Chemistry**

In which of the following reactions the hydrogen peroxide acts as a reducing agent?

(A) $Mn^{2+} + H_2O_2 \rightarrow Mn^{4+} + 2OH^-$

(B) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(C) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(D) $2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2OH^-$

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

The metal which is extracted by oxidation and subsequent reduction from its ore is :

(A) Al (B) Cu (C) Ag (D) Fe

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Beryllium has less negative value of reduction potential compared to the other alkaline earth metals.

Reason R : Beryllium has large hydration energy due to small size of Be^{2+} but relatively large value of atomization enthalpy

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) Both A and R are correct and R is the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Correct statement is :

(A) An average human being consumes more food than air

(B) An average human being consumes 100 times more air than food

(C) An average human being consumes nearly 15 times more air than food

(D) An average human being consumes equal amount of food and air

Open solution

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Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Identify the correct statements about alkali metals.

A. The order of standard reduction potential ($M^+ | M$) for alkali metal ions is Na > Rb > Li.

B. CsI is highly soluble in water.

C. Lithium carbonate is highly stable to heat.

D. Potassium dissolved in concentrated liquid ammonia is blue in colour and paramagnetic.

E. All the alkali metal hydrides are ionic solids.

Choose the correct answer from the options given below :

(A) C and E only **(B)** A and E only **(C)** A, B, D only **(D)** A, B and E only

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List-I Type		List-II Name
A.	Antifertility drug	i)	Norethindrone
B.	Tranquilizer	ii)	Meprobomate
C.	Antihistamine	iii)	Seldane
D.	Antibiotic	iv)	Ampicillin

Choose the correct answer from the options given below :

(A) A-I, B-III, C-II, D-IV **(B)** A-I, B-II, C-III, D-IV

(C) A-II, B-I, C-III, D-IV **(D)** A-IV, B-III, C-II, D-I

Q80. Numerical · **Chemistry**

Open solution

The number of statement/s which are the characteristics of physisorption is _____

A. It is highly specific in nature

B. Enthalpy of adsorption is high

C. It decreases with increase in temperature

D. It results into unimolecular layer

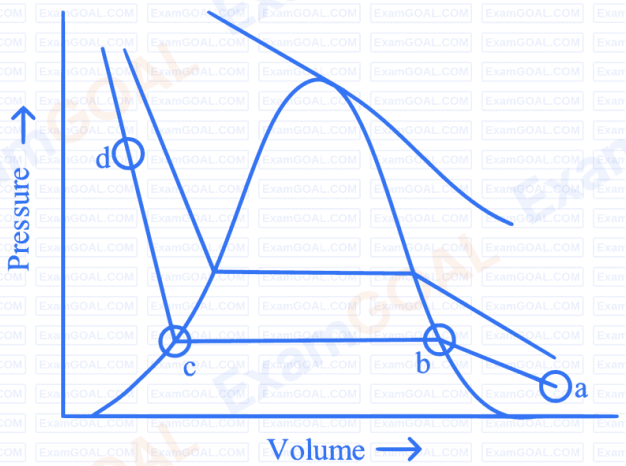
E. No activation energy is needed

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Q81. Numerical · **Chemistry**

Open solution

The number of statement/s, which are correct with respect to the compression of carbon dioxide from point (a) in the Andrews isotherm from the following is _____



A. Carbon dioxide remains as a gas upto point (b)

B. Liquid carbon dioxide appears at point (c)

C. Liquid and gaseous carbon dioxide coexist between points (b) and (c)

D. As the volume decreases from (b) and (c), the amount of liquid decreases

Q82. MCQ (Single Correct Answer) · **Mathematics**

Open solution

If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to :

(A) 3 (B) 1 (C) -1 (D) 5

Q83. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let A be a 3 $\times$ 3 matrix such that $|\text{adj}(\text{adj}(\text{adj A}))| = 12^4$. Then $|\text{A}^{-1}\ \text{adj A}|$ is equal to

(A) 12 (B) $2\sqrt{3}$ (C) 1 (D) $\sqrt{6}$

Q84. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let the plane containing the line of intersection of the planes

$P_1 : x + (\lambda + 4)y + z = 1$ and

$P_2 : 2x + y + z = 2$

pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P_2 is :

(A) $2\sqrt{6}$ (B) $3\sqrt{6}$ (C) $4\sqrt{6}$ (D) $5\sqrt{6}$

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Q85. MCQ (Single Correct Answer) · **Mathematics**

The equations of the sides AB and AC of a triangle ABC are $(\lambda+1)x+\lambda y=4$ and $\lambda x+(1-\lambda)y+\lambda=0$ respectively. Its vertex A is on the y-axis and its orthocentre is (1, 2). The length of the tangent from the point C to the part of the parabola $y^2=6x$ in the first quadrant is :

(A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{6}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let p and q be two statements. Then $\sim(p\wedge(p\Rightarrow\sim q))$ is equivalent to

(A) $(\sim p)\vee q$ (B) $p\vee(p\wedge(\sim q))$

(C) $p\vee(p\wedge q)$ (D) $p\vee((\sim p)\wedge q)$

Open solution

Q87. MCQ (Single Correct Answer) · **Mathematics**

If $({}^{30}C_1)^2+2({}^{30}C_2)^2+3({}^{30}C_3)^2+\ldots+30({}^{30}C_{30})^2=\frac{\alpha 60!}{(30!)^2}$ then α is equal to :

(A) 30 (B) 10 (C) 15 (D) 60

Open solution

Q88. Numerical · **Mathematics**

Let $S=\{\theta\in[0,2\pi):\tan(\pi\cos\theta)+\tan(\pi\sin\theta)=0\}$. Then $\sum_{\theta\in S}\sin^2\left(\theta+\frac{\pi}{4}\right)$ is equal to _____.

Open solution

Q89. Numerical · **Mathematics**

If $\frac{1^3+2^3+3^3+\ldots\ up\ to\ n\ terms}{1\,.\,3+2\,.\,5+3\,.\,7+\ldots\ up\ to\ n\ terms}=\frac{9}{5}$, then the value of n is

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

Match List I with List II

	List-I		List-II
A.	AM Broadcast	I.	88-108 MHz
B.	FM Broadcast	II.	540-1600 kHz
C.	Television	III.	3.7-4.2 GHz
D.	Satellite Communication	IV.	54MHz - 890MHz

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-I, D-II (B) A-II, B-I, C-IV, D-III

(C) A-I, B-III, C-II, D-IV (D) A-II, B-III, C-I, D-IV

Open solution

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Answer Key

Chemistry

1. **A**	2. **B**	3. **C**	4. **B**	5. **C**
6. **C**	7. **C**	8. **A**	9. **A**	10. **C**
11. **A**	12. **A**	13. **A**	14. **B**	15. **8**
16. **620**	17. **3**	18. **2**	19. **8**	20. **5**
21. **314**	22. **85**			

Mathematics

23. **B**	24. **C**	25. **C**	26. **B**	27. **D**
28. **D**	29. **A**	30. **D**	31. **D**	32. **D**
33. **B**	34. **B**	35. **A**	36. **D**	37. **13**
38. **384**	39. **8**	40. **405**	41. **122**	42. **27**
43. **36**	44. **432**			

Physics

45. **C**	46. **B**	47. **C**	48. **A**	49. **D**
50. **D**	51. **B**	52. **D**	53. **B**	54. **C**
55. **B**	56. **A**	57. **D**	58. **A**	59. **A**
60. **B**	61. **B**	62. **D**	63. **B**	64. **54**
65. **100**	66. **6**	67. **32**	68. **44**	69. **105**
70. **9**	71. **7**	72. **1**	73. **3**	

Out of syllabus

74. **B**	75. **C**	76. **B**	77. **C**	78. **B**
79. **B**	80. **2**	81. **2**	82. **D**	83. **B**
84. **B**	85. **B**	86. **A**	87. **C**	88. **2**
89. **5**	90. **B**			

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